

Lecture 11: Maximum Likelihood Method, Logistic Regression, and Generalized linear model

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September 28, 2026



- 1 Maximum likelihood method for linear regression
- 2 Logistic Regression for Binary Response
 - Regression models with binary response variable
 - Logistic response functions for binary responses
 - Maximum likelihood estimation
 - Logistic regression
 - Inferences about regression parameters
- 3 Generalized linear model

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Normal error regression model - simple linear regression

The **normal error regression model** is as follows:

$$Y_i = \beta_0 + \beta_1 X_i + \epsilon_i, \quad i = 1, \dots, n$$

where

- Y_i is the observed response in the i th trial
- X_i is constant, the level of the predictor variable in the i th trial
- β_0 and β_1 are parameters
- $\epsilon_i, i = 1 \dots n$ are identical, independent, normal distribution $N(0, \sigma^2)$
- Hence, Y_i is a normal distribution with mean $\beta_0 + \beta_1 X_i$ and variance σ^2 .

- From the assumption about of normal error regression model, the likelihood value for all Y_i is given by

$$\begin{aligned} L(\beta_0, \beta_1, \sigma) &= \prod_{i=1}^n \frac{1}{(2\pi\sigma^2)^{1/2}} \exp \left[-\frac{1}{2\sigma^2} (Y_i - \beta_0 - \beta_1 X_i)^2 \right] \\ &= \frac{1}{(2\pi\sigma^2)^{n/2}} \exp \left[-\frac{1}{2\sigma^2} \sum_{i=1}^n (Y_i - \beta_0 - \beta_1 X_i)^2 \right] \end{aligned}$$

Likelihood is a function of $\sigma, \beta_i, i = 0, \dots, p - 1..$

Maximum likelihood estimators vs. least-square estimators

- $\hat{\beta}_0$, $\hat{\beta}_1$, and $\hat{\sigma}^2$ are the maximum likelihood estimator of β_0 , β_1 , and σ^2 .
- It is proved that maximum likelihood estimators $\hat{\beta}_0$, $\hat{\beta}_1$ are also the least-square estimator b_0 , b_1 founded previously

$$b_1 = \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{\sum_{i=1}^n (X_i - \bar{X})^2}$$

$$b_0 = \frac{1}{n} \left(\sum_{i=1}^n Y_i - b_1 \sum_{i=1}^n X_i \right) = \bar{Y} - b_1 \bar{X}.$$

- Maximum likelihood estimator of the variance is

$$\hat{\sigma}^2 = \frac{\sum (Y_i - \hat{Y}_i)^2}{n},$$

which is an biased estimator of variance.

- Note that the unbiased estimator from least square method is $MSE = \frac{\sum (Y_i - \hat{Y}_i)^2}{n-p}$. These two estimators are similar if n is large.

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Regression models with binary response variable

- Consider the simple regression model:

$$Y_i = \beta_0 + \beta_1 X_i + \epsilon_i,$$

where the outcome Y_i is binary, taking on the value of either 0 or 1.

- Assume that Y_i is a Bernoulli random variable with the probability:
 $P(Y_i = 1) = \pi_i$ and $P(Y_i = 0) = 1 - \pi_i$.
- Hence,

$$\begin{aligned} E\{Y_i\} &= \beta_0 + \beta_1 X_i \\ &= 1(\pi_i) + 0(1 - \pi_i) = \pi_i. \end{aligned}$$

Logistic mean response function

- Using logistic function to model π_i

$$E\{Y_i\} = \pi_i = \frac{\exp(\beta_0 + \beta_1 X_i)}{1 + \exp(\beta_0 + \beta_1 X_i)} = \frac{1}{1 + \exp(-\beta_0 - \beta_1 X_i)}$$

or in vector form

$$E\{Y_i\} = \pi = \frac{\exp(X_i' \beta)}{1 + \exp(X_i' \beta)} = \frac{1}{1 + \exp(-X_i' \beta)}$$

- Let $a_i = \beta_0 + \beta_1 X_i$ ($a_i = X_i' \beta$ in vector form), we have $\pi_i = \frac{\exp(a_i)}{1 + \exp(a_i)}$.
- Hence,

$$a_i = \ln \frac{\pi_i}{1 - \pi_i},$$

which is called the **logit transformation** of the probability π_i .

- The ratio $\frac{\pi_i}{1 - \pi_i}$ is called the **odds**.
- Logit transformation is also called **log-odds**.

- Error term: $\epsilon_i = 1 - (\beta_0 + \beta_1 X_i)$ if $Y_i = 1$, and $\epsilon_i = -(\beta_0 + \beta_1 X_i)$ if $Y_i = 0$.

Errors in the logistic model is **not normal**.

- Error variance: $\sigma\{\epsilon_i\} = \pi_i(1 - \pi_i)$.

Error variance in the logistic model is **not constant**.

Likelihood function

- The probability that $Y_i = 1$ or 0 is

$$f_i(Y_i) = \pi_i^{Y_i}(1 - \pi_i)^{(1-Y_i)}$$

- Since the Y_i observations are independent, their joint probability function is (likelihood):

$$g(Y_1, Y_2, \dots, Y_n) = \prod_{i=1}^n f_i(Y_i) = \prod_{i=1}^n \pi_i^{Y_i}(1 - \pi_i)^{(1-Y_i)}$$

- Log-likelihood:

$$\ln g(Y_1, Y_2, \dots, Y_n) = \sum_{i=1}^n \left[Y_i \ln \left(\frac{\pi_i}{1 - \pi_i} \right) \right] + \sum_{i=1}^n \ln(1 - \pi_i)$$

Maximum likelihood estimation

- Since $\pi_i = \frac{\exp(\beta_0 + \beta_1 X_i)}{1 + \exp(\beta_0 + \beta_1 X_i)}$, log-likelihood is rewritten as:

$$\ln L(\beta_0, \beta_1) = \sum_{i=1}^n Y_i(\beta_0 + \beta_1 X_i) - \sum_{i=1}^n \ln [1 + \exp(\beta_0 + \beta_1 X_i)].$$

- We want to find β_0 and β_1 to maximize the log-likelihood.
- No closed-form solution exists.
- We solve $\max_{\beta_0, \beta_1} \ln L(\beta_0, \beta_1)$ by iterative algorithms like gradient decent, stochastic gradient descent, mini-batch gradient descent, *etc.*

Example

Please see the notebook.

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Multiple logistic regression

- We replace $\beta_0 + \beta_1 X_i$ by

$$\beta_0 + \beta_1 X_{1,i} + \dots + \beta_{p-1} X_{p-1,i} = X_i' \beta.$$

- Multiple logistic response function is

$$E\{Y_i\} = \frac{\exp(X_i' \beta)}{1 + \exp(X_i' \beta)} = \frac{1}{1 + \exp(-X_i' \beta)}$$

- Logit response function or linear predictor is:

$$\ln \left(\frac{\pi_i}{1 - \pi_i} \right) = X_i' \beta.$$

- Log-likelihood function is:

$$\ln L(\beta) = \sum_{i=1}^n Y_i (X_i' \beta) - \sum_{i=1}^n \ln [1 + \exp(X_i' \beta)].$$

Test concerning a single β_k : Wald test

- $H_0 : \beta_k = 0, H_a : \beta_k \neq 0$.
- Test statistic is $z^* = \frac{b_k}{s\{b_k\}}$.
- The decision rule is:
 - If $|z^*| \leq z(1 - \alpha/2)$, conclude H_0
 - If $|z^*| > z(1 - \alpha/2)$, conclude H_a

Interval estimation of a single β_k

- $1 - \alpha$ confidence limits for β_k is:

$$b_k \pm z(1 - \alpha/2) * s\{b_k\}.$$

Programming example

Please see the notebook of programming task example.

Test whether several $\beta_k = 0$: Likelihood ratio test

- Alternatives:

- $H_0 : \beta_q = \beta_{q+1} = \dots = \beta_{p-1} = 0$,
- H_a : not all of the β_k in H_0 equal zero.

- The reduced logistic model:

$$\pi = [1 + \exp(-X'\beta_R)]^{-1}$$

where $X'\beta_R = \beta_0 + \beta_1 X_1 + \dots + \beta_{q-1} X_{q-1}$.

- Test statistic:

$$G^2 = -2 \ln \left(\frac{L(R)}{L(F)} \right),$$

where $L(R)$ and $L(F)$ are likelihood function of reduced model and full model, respectively.

- Decision rule:

- If $G^2 \leq \chi^2(1 - \alpha; p - q)$, conclude H_0
- If $G^2 > \chi^2(1 - \alpha; p - q)$, conclude H_a

Programming example

Please see the notebook of Disease Outbreak example.

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Generalized linear model

The class of generalized linear models can be described as follows:

- $Y_1, \dots, Y_n$ are n independent responses that follow the exponential family of distributions, e.g., the normal, binomial, Poisson, inverse normal, exponential, and gamma distributions, with expected value:

$$E\{Y_i\} = \mu_i.$$

- A linear predictor based on the predictor variables $X_{i1}, \dots, X_{i,p-1}$:

$$X_i' \beta = \beta_0 + \beta_1 X_{i1} + \dots + \beta_{p-1} X_{i,p-1}.$$

- The link function g relates the linear predictor to the mean response:

$$X_i' \beta = g(\mu_i).$$

Generalized linear model - link functions

- For linear regression:

$$g(\mu_i) = X_i' \beta.$$

- For logistic regression:

$$g(\mu_i) = g(\pi_i) = \ln \left(\frac{\pi_i}{1 - \pi_i} \right) = X_i' \beta$$

- See GLM, statsmodels <https://www.statsmodels.org/dev/examples/notebooks/generated/glm.html>

References

- Chapter 13, Montgomery *et.al.*'s book
- Chapter 14, Kutner *et.al.*'s book

Thank you.